

Effective-Orbit Quotients of Frozen First-Passage Laws

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Abstract

We quotient the exact twenty-target first-passage atlas by the faithful order-1920 label action inherited from the frozen closure model. The ambient lifted order 11520 is kept distinct. Rebuilding the five named label permutations and checking their action on all 65536 support indicators produces sixteen target orbits: twelve singletons and four doubletons. The C88 first-passage laws and the C92 pivotal-label rank rows are transported equivariantly at generator level. This is a finite quotient result under `NO_BAD_EULER_OR_ROOT_NUMBER`.

1 Two group orders

The C75 lifted closure-incidence symmetry has order 11520. Its order-six kernel on labels is removed before forming the faithful effective label image, which has order 1920. C93 uses only this effective image, generated by the five frozen permutations `zero_5_cycle`, `zero_transposition`, `fiber_F3_transposition`, `fiber_F9_transposition`, and `ambient_s`.

2 Target transport

For a support A , the C88 indicator records whether the generated closure contains H . Applying a named-label permutation transports every support indicator. Reconstructing the maximal contained target from each transformed support gives a permutation of the twenty actual targets. Theorem-level checks on each of the five generators show that the complete first-passage count vector is preserved under this transport.

Theorem 1. *The effective action partitions the twenty target laws into sixteen orbits, with orbit-size spectrum $12 \times 1 + 4 \times 2$. For every generator g ,*

$$\Pr(T_H = k) = \Pr(T_{gH} = k),$$

and the pivotal-rank row for label S_i at H equals the row for gS_i at gH . The C88 support indicators and C92 sensitivity rows satisfy these identities exactly.

3 Audit and scope

The independent checker rebuilds the source-bound permutations and all target maps, then verifies the C92 rank rows. SymPy checks every orbit generating polynomial and its total mass and mean. Clean replay preserves the evidence hash

`4104f181b88d83666c9fcff814a7029a148c498e6393ad181c60fe5133adb9fe`.

No arithmetic/local data, Euler factors, root numbers, automorphy, full Burnside ring or table of marks, or Hilbert–Polya operator is claimed.